



SIMATS ENGINEERING



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ASSESSMENT TOOL -2

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INDUSTRY PROBLEM-SOLVING TASK

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan.

The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

ANSWER :-

Network Topology for a 20-Person Startup

Introduction

A startup with 20 employees requires a network that is reliable, easy to manage, and capable of supporting future growth. Among the different network topologies, the star topology is the most suitable because it offers high performance, simple troubleshooting, and easy expansion.

Recommended Network Design

In a star topology, every device is connected to a central managed switch. A Layer 2/Layer 3 managed switch, such as a Cisco Catalyst or Netgear ProSAFE, with at least 24 ports is recommended. This provides enough connections for all employee computers, printers, VoIP phones, and additional devices while allowing room for future expansion.

Each workstation is connected to the switch using Cat6 UTP cables. These cables support Gigabit Ethernet speeds (1 Gbps) over distances of up to 100 meters, making them ideal for a standard office environment.

The central switch is connected to a router/firewall, such as a Cisco RV Series router or a pfSense appliance. The router manages internet connectivity, Dynamic Host Configuration

Protocol (DHCP), Network Address Translation (NAT), and network security. A wireless access point (Wi-Fi AP) is also connected to the switch to provide wireless connectivity for laptops, smartphones, and tablets.

Advantages of the Star Topology

The star topology provides several benefits for a small business:

- **Easy Troubleshooting:** Network problems can be quickly identified because each device has its own dedicated connection to the switch.
- **Fault Isolation:** If one cable or workstation fails, only that device is affected, while the remaining employees continue working without interruption.
- **Scalability:** New computers, printers, or other devices can be added easily without affecting the existing network.
- **Centralized Management:** The managed switch allows administrators to monitor network traffic, configure security settings, and improve overall performance.
- **High Performance:** Dedicated connections reduce data collisions and ensure reliable Gigabit network speeds.

Conclusion

A star topology is the best networking solution for a 20-person startup because it is reliable, scalable, and easy to manage. Using a 24-port managed switch, Cat6 UTP cabling, a router/firewall, and a wireless access point creates a secure and efficient network that supports daily business operations while allowing for future growth.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

ANSWER :-

Network Topology for a Bank Requiring Near-Zero Downtime

Introduction

Banks require highly reliable and secure network infrastructure because financial transactions must be available at all times. Even a short period of network downtime can result in financial losses and affect customer trust. Therefore, a full mesh or partial mesh topology is the most suitable network design for a banking environment.

Recommended Network Design

In a full mesh topology, every core network device, including routers, core switches, and servers, is directly connected to every other device. This provides multiple communication paths and eliminates a single point of failure.

In practice, most banks implement a partial mesh topology to reduce costs while maintaining high availability. The core layer consists of interconnected core switches and firewalls, while the distribution and access layers connect to at least two core switches through redundant uplinks. This design ensures continuous network connectivity even if one connection fails.

The network uses dual enterprise-grade routers, such as Cisco ASR or Juniper MX Series, to provide redundant internet connectivity using the Border Gateway Protocol (BGP). Connections between core devices are established using fiber optic cables. Single-mode fiber is used for long-distance communication between buildings, while multi-mode OM4 fiber is used within the data center. Fiber optic cabling provides high bandwidth, low latency, and immunity to electromagnetic interference, making it ideal for banking operations.

To improve reliability, the network also includes redundant power supplies, Uninterruptible Power Supply (UPS) systems, and RAID storage arrays, ensuring continuous operation during hardware or power failures.

Advantages of the Mesh Topology

The mesh topology provides several important benefits for a banking network:

- **High Availability:** Multiple communication paths ensure continuous network operation.
- **Fault Tolerance:** If a network link or device fails, traffic is automatically rerouted through another available path.
- **Fast Recovery:** Routing protocols such as OSPF and BGP quickly detect failures and redirect traffic within milliseconds.
- **Improved Security and Reliability:** Redundant network components reduce the risk of service interruption.
- **Scalability:** Additional network devices can be integrated without significantly affecting network performance.

Conclusion

A full mesh or partial mesh topology is the most appropriate solution for a banking network because it provides maximum reliability, fault tolerance, and continuous availability. By using redundant routers, fiber optic connections, backup power systems, RAID storage, and dynamic routing protocols, the network can maintain uninterrupted financial services while meeting strict regulatory requirements for uptime, security, and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal

cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely

acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

ANSWER :-

Network Topology for a Small Classroom Lab with a Limited Budget

Introduction

A small classroom computer lab requires a network that is affordable, simple to install, and easy to maintain. Since the budget is limited, the bus topology is a suitable choice because it minimizes hardware costs while providing basic network connectivity for classroom activities.

Recommended Network Design

In a bus topology, all computers are connected to a single communication cable known as the backbone. A traditional bus network uses an RG-58 (50-ohm) coaxial cable, where each computer is connected through a T-connector and a BNC connector. 50-ohm terminators are installed at both ends of the cable to prevent signal reflection and ensure proper communication.

A modern and more practical variation uses an 8-port or 16-port unmanaged switch with Cat5e or Cat6 UTP cables to connect each computer. This approach provides similar low-cost

networking while offering better performance, easier installation, and improved reliability compared to the traditional coaxial bus network.

The classroom does not require a dedicated server. Instead, the teacher's computer can share files, printers, and other resources using the built-in file-sharing features available in Windows or Linux operating systems.

Advantages of the Bus Topology

The bus topology offers several benefits for a classroom environment:

- **Low Cost:** Requires minimal networking hardware, making it an economical solution.
- **Simple Installation:** Easy to set up and suitable for small networks.
- **Easy Resource Sharing:** Computers can share files and printers without requiring a dedicated server.
- **Suitable for Small Networks:** Meets the networking needs of classrooms with light internet browsing and file-sharing activities.

Limitations

Despite its low cost, the bus topology has some disadvantages:

- **Single Point of Failure:** If the main communication cable is damaged, the entire network stops functioning.
- **Data Collisions:** Multiple devices transmitting simultaneously can reduce network performance.
- **Limited Scalability:** Performance decreases as more computers are added to the network.

Conclusion

The bus topology is a cost-effective networking solution for a small classroom laboratory. It provides basic connectivity with minimal hardware requirements, making it suitable for educational environments with limited budgets. Although it has limitations such as cable dependency and data collisions, it remains an appropriate choice for classrooms with light network usage.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

ANSWER :-

Network Topology for a Large University Campus

Introduction

A large university campus consists of multiple buildings, including lecture halls, laboratories, libraries, hostels, and administrative offices. Since thousands of students, faculty, and staff access the network simultaneously, a single network topology cannot efficiently meet all requirements. Therefore, a hybrid topology is the most suitable solution because it combines the strengths of different topologies to provide high performance, reliability, and scalability.

Recommended Network Design

A hybrid topology combines ring, mesh, star, and tree topologies to create a flexible and efficient campus network.

At the campus backbone, single-mode fiber optic cables supporting 10 Gbps or 40 Gbps speeds connect the core routers and multilayer switches in a ring or partial mesh topology. This ensures redundancy so that if one fiber link fails, data is automatically rerouted through another path without interrupting network services.

Each campus building is connected to the backbone through a distribution switch using dual fiber optic uplinks for increased reliability. Inside every building, a star topology is implemented, where floor-level access switches connect computers, printers, and other network devices using Cat6 UTP cables.

Wireless connectivity is provided through Wi-Fi access points connected to the access switches using Power over Ethernet (PoE). PoE allows both power and data to be transmitted through a single Ethernet cable, reducing installation costs and eliminating the need for separate power cables.

Advantages of the Hybrid Topology

The hybrid topology provides several benefits for a university campus:

- **High Scalability:** New buildings and departments can be added easily without disrupting the existing network.
- **High Reliability:** Redundant fiber links ensure continuous network availability.
- **Efficient Network Management:** The hierarchical design makes troubleshooting and maintenance easier.
- **High Performance:** Fiber optic backbone connections provide fast and low-latency communication across the campus.
- **Flexible Design:** Different topologies are used where they are most effective, improving overall network efficiency.

Conclusion

A hybrid topology is the best networking solution for a large university campus because it combines the advantages of multiple network designs. By using a fiber optic backbone, star topology within buildings, and a three-layer hierarchical architecture, the network provides excellent performance, redundancy, scalability, and simplified management. This design meets the communication needs of thousands of users while supporting future campus expansion.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

ANSWER :-

Network Design for a Company with Multiple Departments

Introduction

A company with multiple departments such as Human Resources (HR), Finance, Sales, and Information Technology (IT) requires a secure, reliable, and scalable network. A hierarchical network design using switches, routers, and VLANs is the most suitable solution because it improves performance, enhances security, and supports future business growth.

Recommended Network Design

Each department is connected to its own Layer 2 switch, which forms the access layer of the network. Devices such as desktop computers, laptops, printers, and IP phones communicate through these switches within their respective departments.

All Layer 2 switches are connected to a Layer 3 switch or router, which acts as the core layer of the network. The Layer 3 device performs inter-VLAN routing, allowing secure communication between different departments when required.

To improve network security and reduce unnecessary broadcast traffic, Virtual Local Area Networks (VLANs) are created for each department. For example, the HR, Finance, Sales, and

IT departments each have their own VLAN. This logical separation protects sensitive information while improving network efficiency.

A router with firewall functionality connects the company's internal network to the internet. The firewall protects the network from unauthorized access and cyber threats. To improve reliability, redundant links and backup network paths can be implemented so that communication continues even if a network connection fails.

The network also uses proper IP addressing and subnetting, making network management easier and ensuring efficient utilization of IP addresses.

Advantages of the Network Design

The proposed network design provides several benefits:

- **Improved Security:** VLANs isolate departments and protect confidential information.
- **Better Performance:** Reduced broadcast traffic improves overall network efficiency.
- **Easy Scalability:** New departments and devices can be added without affecting the existing network.
- **High Reliability:** Redundant links and backup paths minimize network downtime.
- **Simplified Network Management:** A hierarchical architecture makes monitoring, troubleshooting, and maintenance easier.

Conclusion

A hierarchical network design using Layer 2 switches, a Layer 3 switch or router, VLANs, and a firewall is the best solution for a company with multiple departments. This design provides secure communication, efficient network management, high performance, and easy scalability, making it suitable for both current operations and future business expansion.